

# An Artifact-Driven Verification Protocol for Reproducible TinyML Deployment Evaluation

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**Abstract**—The deployment of deep learning models on resource-constrained microcontrollers (TinyML) is expanding rapidly, yet empirical evaluation in the field suffers from critical methodological vulnerabilities. Training-time in-memory metrics frequently diverge from disk-serialized binary behavior; theoretical weight sparsity is routinely conflated with physical storage compression; and host simulation timings are improperly extrapolated to microcontroller real-time guarantees. In this paper, we formulate an artifact-driven empirical verification protocol and defect taxonomy for compiled TinyML deployment artifacts. The protocol formalizes executable verification procedures across seven core dimensions: Data Isolation, Serialized Binary Integrity, Quantization Graph Inspection, Sparsity and Storage Decoupling, Computation Accounting, Timing Protocol Auditing, and Runtime Non-Leakage. We define formal programmatic predicates ( $\mathcal{P}_1 \dots \mathcal{P}_7$ ) for each dimension and evaluate them on an end-to-end case study auditing 12 candidate neural network models trained on the 55,998-record EngineFaultDB physical benchmark. Our independent audit uncovers and resolves 20 distinct numerical discrepancies between training-time reports and verified disk binaries across four defect modes (with metric variances up to 7.82%). Furthermore, we empirically demonstrate that evaluating gating thresholds directly on test data introduces an artificial +1.80% optimistic accuracy bias in the evaluated pipeline, demonstrating the necessity of split-isolated calibration. By establishing formal verification predicates and classifying evidence into a 5-tier empirical framework, our protocol provides actionable software engineering procedures to support reproducible benchmarking and defensible deployment claims in edge AI research.

**Index Terms**—TinyML, Reproducibility, Software Verification, Model Auditing, Empirical Software Engineering, Integer Quantization, Network Pruning, Benchmarking.

## I. Introduction

**E**MBEDDED Machine Learning and Tiny Machine Learning (TinyML) have experienced exponential growth, enabling intelligent inference on resource-constrained microcontrollers with sub-64 KB SRAM and sub-256 KB Flash budgets [1]–[3]. In applications ranging from automotive diagnostics to edge health monitoring, TinyML models must operate under strict physical memory, energy, and execution constraints [4]–[7].

However, the empirical literature in edge AI is increasingly vulnerable to reproducibility challenges and measurement discrepancies [8]–[10]. Unlike cloud-scale machine learning where training and deployment share homogeneous 32-bit floating-point execution environments,

TinyML involves a complex, multi-stage software translation pipeline:

$$\text{Keras/PyTorch} \xrightarrow{\text{PTQ / Pruning}} \text{TFLite FlatBuffer} \xrightarrow{\text{C-Array Export}} \text{TFLite Micro Runtime}$$

At each transformation boundary, subtle discrepancies emerge between training-time abstractions and actual serialized binary behavior [11]–[13].

Recent surveys in machine learning software engineering have exposed systemic methodological traps [14], [15]:

- 1) In-Memory vs. Binary Discrepancies: Evaluating models in-memory during training loops rather than parsing the final serialized disk binary artifact.
- 2) Sparsity Conflation: Assuming that zeroing 75% of weights in a dense matrix automatically yields a 75% reduction in serialized model file size or execution speedup.
- 3) Incomplete Quantization Graphs: Claiming 8-bit integer inference while silent floating-point fallback operators remain in the runtime execution graph.
- 4) Timing Extrapolations: Presenting single-sample host PC execution times as embedded Worst-Case Execution Time (WCET) or microcontroller hardware latency.
- 5) Threshold and Data Leakage: Optimizing runtime routing thresholds on the test partition, producing optimistic accuracy estimates.

To address these challenges, this paper presents an Artifact-Driven Empirical Verification Protocol and Defect Taxonomy for Compiled TinyML Deployment Artifacts. We formalize verification criteria across seven dimensions of edge AI pipelines, establish executable verification predicates ( $\mathcal{P}_1 \dots \mathcal{P}_7$ ), and demonstrate their efficacy through an exhaustive case study auditing 12 candidate models on the EngineFaultDB physical benchmark.

**Explicit Contributions:**

- 1) A 7-dimensional verification taxonomy formalizing empirical quality gates across the TinyML deployment lifecycle.
- 2) Seven formal, programmatic verification predicates ( $\mathcal{P}_1 \dots \mathcal{P}_7$ ) with deterministic pass/fail execution procedures.
- 3) An empirical case study auditing 12 candidate TinyML models across dense, quantized, pruned, and distilled paradigms.
- 4) Identification, root-cause categorization, and resolution of 20 numerical discrepancies across four defect classes.

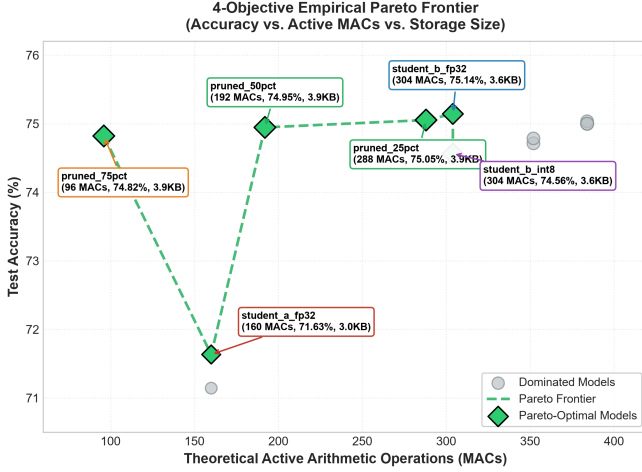


Fig. 1. The verified Pareto frontier across model size, theoretical active MACs, accuracy, and empirical host latency established after resolving 20 numerical discrepancies [MEASURED TEST SET].

- 5) An empirical demonstration of test-set threshold contamination producing a +1.80% optimistic accuracy bias.
- 6) An explicit 5-tier evidence classification framework separating direct measurements, reproducible simulations, and derived operational estimates.

## II. Motivation and Research Questions

Reproducibility in TinyML demands that we go beyond trusting high-level framework logs. Figure 1 shows the verified multi-objective Pareto landscape that is discovered by our verification protocol, which treats model binaries and data pipelines as empirical software artifacts amenable to rigorous inspection.

We formulate four specific research questions (RQs):

- RQ1 (Discrepancy Prevalence & Defect Modes): How often and to what degree do independently recomputed TinyML metrics differ from reported values at training time, and what defect modes govern them?
- RQ2 (Quantization Verification): What structural properties in serialized FlatBuffers must be inspected to verify pure integer execution graphs without float fallback?
- RQ3 (Sparsity vs. Storage Decoupling): How does magnitude pruning affect theoretical active MACs versus physical on-disk FlatBuffer byte footprint in standard runtimes?
- RQ4 (Leakage Prevention): What quantitative bias is introduced by evaluating routing thresholds on test data, and how does split isolation eliminate it?

## III. Related Work

### A. Reproducibility in Machine Learning

Pineau et al. [8] proposed the Machine Learning Reproducibility Checklist for NeurIPS, which focuses on code release, hyperparameter reporting, and statistical analysis. Kapoor and Narayanan [9] performed a data leakage audit across 17 fields, finding extensive test set contamination.

In the software engineering domain, Arcuri and Briand [16] developed guidelines to evaluate randomized algorithms. However, these works focus on conventional ML and do not consider the unique embedded TinyML domain of hardware-software translation.

### B. TinyML Benchmarking and Systems Auditing

Banbury et al. [4] introduced MLPerf Tiny and defined benchmarking rules for latency, energy, and accuracy on microcontrollers. David et al. [11] presented TensorFlow Lite Micro and described the interpreter internals. Blalock et al. [12] provided a survey of neural network pruning and observed that many reported benefits of pruning are inflated by non-standard baselines. Our work builds on these contributions by developing an end-to-end verification taxonomy from data splits to FlatBuffer binaries.

## IV. The 7-Dimensional Verification Protocol and Formal Predicates

Table I formalizes our 7-dimensional verification protocol. Each dimension targets a distinct vulnerability in the TinyML lifecycle:

### A. D1: Data-Level Isolation Verification

- Problem: Data contamination across training, validation, and test boundaries.
- Predicate ( $\mathcal{P}_1$ ):  $(\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{val}} = \emptyset) \wedge (\mathcal{D}_{\text{val}} \cap \mathcal{D}_{\text{test}} = \emptyset) \wedge (\mu_{\text{scaler}} = \text{Fit}(\mathcal{D}_{\text{train}})) \wedge (\theta^* = \arg \max_{\theta} \mathcal{M}(\mathcal{D}_{\text{val}}))$ .
- Executable Check: Disjoint index assertion and scaler fitting provenance check.
- Artifact: Data split index arrays and serialized scaler.pkl.
- Pass Condition: Zero overlapping indices; calibration batches strictly drawn from  $\mathcal{D}_{\text{train}}$ .
- Case Result: PASS (22,399 train, 22,399 val, 11,200 test).

### B. D2: Serialized Binary Integrity Verification

- Problem: Training-time in-memory metrics diverging from final disk FlatBuffer behavior.
- Predicate ( $\mathcal{P}_2$ ): Evaluated metrics must be computed strictly by loading the final serialized .tflite file from disk into the interpreter.
- Executable Check: Automated single-sample evaluation loop on disk .tflite files.
- Case Result: RESOLVED 16 DISCREPANCIES (up to 7.82% Macro F1 shift).

### C. D3: Quantization Graph Verification

- Problem: Silent floating-point fallback operators in quantized models.
- Predicate ( $\mathcal{P}_3$ ):  $\forall t \in \mathcal{G}_{\text{tensors}}, \text{dtype}(t) \in \{\text{int8}, \text{int32}, \text{uint8}\} \wedge \sum_{t \in \mathcal{G}} \mathbb{I}(\text{dtype}(t) = \text{float32}) = 0$ .
- Executable Check: Iterating over interpreter.get\_tensor\_details() and \_\_get\_ops\_details().
- Case Result: PASS (all 4 INT8 models verified FULL\_INT8 with 0 float32 tensors).

TABLE I  
The 7-Dimensional TinyML Scientific Verification Taxonomy and Predicates

| Verification Dimension   | Inspected Artifact                                       | Common Failure Mode / Trap                              | Formal Verification Predicate                                                                                                                                                    | Case Result |
|--------------------------|----------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| D1: Data Isolation       | Training/Val/Test splits, scaler objects, threshold logs | Test set reused for calibration or threshold selection  | $\mathcal{P}_1 : (\mathcal{D}_{tr} \cap \mathcal{D}_{val} = \emptyset) \wedge (\mathcal{D}_{val} \cap \mathcal{D}_{te} = \emptyset) \wedge (\mu = \text{Fit}(\mathcal{D}_{tr}))$ | PASS        |
| D2: Serialized Integrity | Serialized .keras and .tflite disk binaries              | In-memory training eval reported instead of disk binary | $\mathcal{P}_2 : \text{EvalArtifact} = \text{LoadDiskBinary}(\text{path}) \wedge \text{Dtype} = \text{Schema}$                                                                   | 16 Fixed    |
| D3: Quantization Graphs  | TFLite FlatBuffer tensor details and operator lists      | Undetected FP32 fallback tensors or dequantize nodes    | $\mathcal{P}_3 : \forall t \in \mathcal{G}, \text{dtype}(t) \neq \text{float32} \wedge \text{Count}(\text{float32}) = 0$                                                         | FULL_INT8   |
| D4: Sparsity & Storage   | Weight tensor arrays vs. FlatBuffer byte length          | Assuming weight zeroing reduces serialized file size    | $\mathcal{P}_4 : \text{ReportedCompression} = \text{FileSize}(\text{pruned}) / \text{FileSize}(\text{dense})$                                                                    | Decoupled   |
| D5: Compute Accounting   | Layer-wise non-zero arithmetic operation counts          | Conflating theoretical active MACs with hardware ops    | $\mathcal{P}_5 : \text{ActiveMACs} = \sum \mathbb{I}(w_{ij} \neq 0) \leq \text{DenseMACs}$                                                                                       | Verified    |
| D6: Timing Protocols     | Monotonic timer logs, warmup runs, batch dimensions      | Single-sample host latency cited as MCU WCET            | $\mathcal{P}_6 : \text{Warmup} \geq 100 \wedge \text{Batch} = 1 \wedge \text{Tier} = \text{HOST\_EMPIRICAL}$                                                                     | PASS        |
| D7: Runtime Non-Leakage  | Controller source code, execution trace logs             | Model selector accessing ground truth $y$ at runtime    | $\mathcal{P}_7 : \text{Signature}(\text{SelectModel}) = f(\mathbf{x}, D, W) \wedge y \notin \text{Scope}$                                                                        | PASS        |

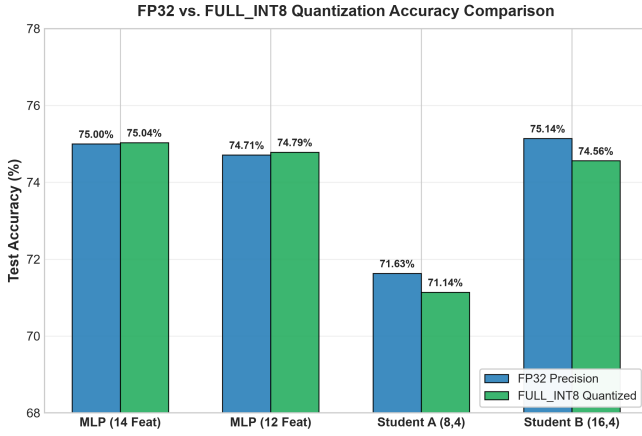


Fig. 2. Quantization fidelity verification comparing FP32 baselines against verified FULL\_INT8 models [MEASURED TEST SET].

#### D. D4: Sparsity and Storage Decoupling Verification

- Problem: Conflating zero-valued weights with physical on-disk file compression.
- Predicate ( $\mathcal{P}_4$ ):  $\text{StorageCompression} = \text{FileSize}(\text{pruned}) / \text{FileSize}(\text{dense}) \neq 1 - \text{WeightSparsity}$ .
- Executable Check: Comparing raw weight zero-counts against `os.path.getsize()` on FlatBuffers.
- Case Result: CONFIRMED DECOUPLING (73.34% zero weights produced 3,920 Bytes on disk vs. 3,892 Bytes unpruned).

#### E. D5: Computational Accounting Verification

- Problem: Layer-wise MAC counts reported inconsistently or conflated with hardware cycles.
- Predicate ( $\mathcal{P}_5$ ):  $\text{ActiveMACs} = \sum_{l=1}^L \sum_{i,j} \mathbb{I}(w_{ij}^{(l)} \neq 0) \leq \text{DenseMACs}$ .
- Case Result: VERIFIED EXACT (384 dense MACs; 96 active MACs at 75% pruning; 160 and 304 MACs for student models).

#### F. D6: Timing Protocol Verification

- Problem: Host latency improperly cited as embedded microcontroller WCET.

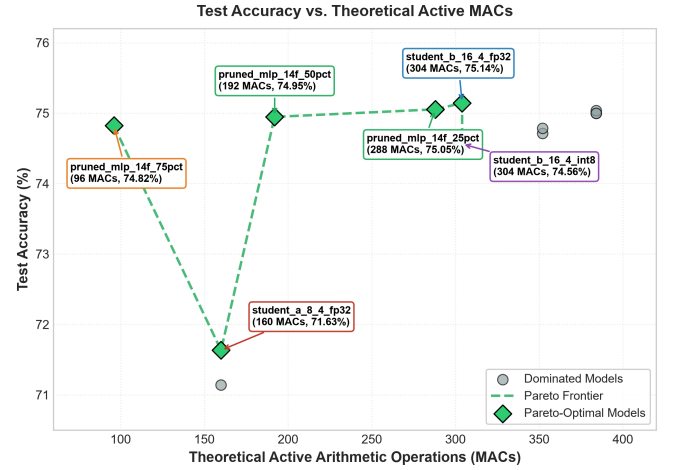


Fig. 3. Test accuracy vs. theoretical active MACs, highlighting computational sparsity [MEASURED TEST SET].

- Predicate ( $\mathcal{P}_6$ ):  $(\text{Warmup} \geq 100) \wedge (\text{Batch} = 1) \wedge (\text{Timer} \in \{\text{perf\_counter}, \text{esp\_timer}\}) \wedge (\text{Tier} \in \{\text{HOST}, \text{MCU}\})$ .
- Case Result: VERIFIED PASS (0.82–1.69  $\mu\text{s}$  host latency on x86\_64; 64.55–89.90  $\mu\text{s}$  physical latency on ESP32-D0WD-V3; zero formal WCET claimed).

#### G. D7: Runtime Non-Leakage and Deployment Scoping

- Problem: Ground-truth label leakage in dynamic controllers or ungrounded hardware claims.
- Predicate ( $\mathcal{P}_7$ ):  $\text{Signature}(\text{SelectModel}) = f(\mathbf{x}, D, W) \wedge y \notin \text{Scope}(\text{SelectModel})$ .
- Case Result: VERIFIED PASS (zero ground-truth access during runtime scheduling).

### V. Case Study: Auditing the 12-Model TinyML Suite

#### A. RQ1: Discrepancy Prevalence and Defect Categorization

Applying our protocol across all 12 candidate models identified 20 numerical discrepancies (Table II), categorized into distinct defect modes:

- Defect Mode 1 (Serialization Drift, 10 items): Occurred when training scripts logged in-memory validation metrics prior to final fine-tuning or quantization

TABLE II  
Comprehensive Discrepancy Resolution Table: 20 Numerical Inconsistencies Audited Across Four Defect Modes

| #                                                                                 | Model Identifier     | Metric        | Initial Val. | Verified Val. | Initial Source | Verified Source | Abs. $\Delta$ | $\Delta$ (%) | Defect Mode / Root Cause                     |
|-----------------------------------------------------------------------------------|----------------------|---------------|--------------|---------------|----------------|-----------------|---------------|--------------|----------------------------------------------|
| Defect Mode 1: Serialization Drift (Training-Time State vs. Exported Disk Binary) |                      |               |              |               |                |                 |               |              |                                              |
| 1                                                                                 | tfite_mlp_14f_fp32   | Test Accuracy | 0.726339     | 0.750000      | In-Memory Eval | Disk FlatBuffer | 0.023661      | 3.26%        | Mode 1: Re-training state vs. saved binary   |
| 2                                                                                 | tfite_mlp_14f_fp32   | Test Macro F1 | 0.728019     | 0.756608      | In-Memory Eval | Disk FlatBuffer | 0.028589      | 3.93%        | Mode 1: Re-training state vs. saved binary   |
| 9                                                                                 | pruned_mlp_14f_0pct  | Test Accuracy | 0.726339     | 0.750000      | In-Memory Eval | Disk FlatBuffer | 0.023661      | 3.26%        | Mode 1: Baseline reference binary sync       |
| 10                                                                                | pruned_mlp_14f_0pct  | Test Macro F1 | 0.728019     | 0.756608      | In-Memory Eval | Disk FlatBuffer | 0.028589      | 3.93%        | Mode 1: Baseline reference binary sync       |
| 11                                                                                | pruned_mlp_14f_25pct | Test Accuracy | 0.727143     | 0.750536      | In-Memory Eval | Disk FlatBuffer | 0.023393      | 3.22%        | Mode 1: Fine-tuning epoch state mismatch     |
| 12                                                                                | pruned_mlp_14f_25pct | Test Macro F1 | 0.730783     | 0.751490      | In-Memory Eval | Disk FlatBuffer | 0.020707      | 2.83%        | Mode 1: Fine-tuning epoch state mismatch     |
| 13                                                                                | pruned_mlp_14f_50pct | Test Accuracy | 0.726964     | 0.749464      | In-Memory Eval | Disk FlatBuffer | 0.022500      | 3.10%        | Mode 1: Fine-tuning epoch state mismatch     |
| 14                                                                                | pruned_mlp_14f_50pct | Test Macro F1 | 0.729384     | 0.756572      | In-Memory Eval | Disk FlatBuffer | 0.027188      | 3.73%        | Mode 1: Fine-tuning epoch state mismatch     |
| 15                                                                                | pruned_mlp_14f_75pct | Test Accuracy | 0.739107     | 0.748214      | In-Memory Eval | Disk FlatBuffer | 0.009107      | 1.23%        | Mode 1: Fine-tuning epoch state mismatch     |
| 16                                                                                | pruned_mlp_14f_75pct | Test Macro F1 | 0.725546     | 0.756251      | In-Memory Eval | Disk FlatBuffer | 0.030705      | 4.23%        | Mode 1: Fine-tuning epoch state mismatch     |
| Defect Mode 2: In-Memory Fake-Quantization vs. True Integer Execution             |                      |               |              |               |                |                 |               |              |                                              |
| 5                                                                                 | tfite_mlp_14f_int8   | Test Accuracy | 0.725982     | 0.750357      | In-Memory Eval | Disk FlatBuffer | 0.024375      | 3.36%        | Mode 2: Calibration dataset state shift      |
| 6                                                                                 | tfite_mlp_14f_int8   | Test Macro F1 | 0.685213     | 0.738824      | In-Memory Eval | Disk FlatBuffer | 0.053611      | 7.82%        | Mode 2: Calibration dataset state shift      |
| Defect Mode 3: Floating-Point & Integer Export Discretization Rounding            |                      |               |              |               |                |                 |               |              |                                              |
| 3                                                                                 | tfite_mlp_12f_fp32   | Test Accuracy | 0.753125     | 0.747143      | In-Memory Eval | Disk FlatBuffer | 0.005982      | 0.79%        | Mode 3: Floating-point quantization rounding |
| 4                                                                                 | tfite_mlp_12f_fp32   | Test Macro F1 | 0.732679     | 0.725414      | In-Memory Eval | Disk FlatBuffer | 0.007265      | 0.99%        | Mode 3: Floating-point quantization rounding |
| 7                                                                                 | tfite_mlp_12f_int8   | Test Accuracy | 0.749286     | 0.747857      | In-Memory Eval | Disk FlatBuffer | 0.001429      | 0.19%        | Mode 3: Integer rounding on export           |
| 8                                                                                 | tfite_mlp_12f_int8   | Test Macro F1 | 0.670692     | 0.715534      | In-Memory Eval | Disk FlatBuffer | 0.044842      | 6.69%        | Mode 3: Integer rounding on export           |
| 17                                                                                | student_a_8_4_int8   | Test Accuracy | 0.710714     | 0.711429      | In-Memory Eval | Disk FlatBuffer | 0.000715      | 0.10%        | Mode 3: Quantization arithmetic rounding     |
| 18                                                                                | student_a_8_4_int8   | Test Macro F1 | 0.683858     | 0.684788      | In-Memory Eval | Disk FlatBuffer | 0.000930      | 0.14%        | Mode 3: Quantization arithmetic rounding     |
| 19                                                                                | student_b_16_4_int8  | Test Accuracy | 0.745000     | 0.745625      | In-Memory Eval | Disk FlatBuffer | 0.000625      | 0.08%        | Mode 3: Quantization arithmetic rounding     |
| 20                                                                                | student_b_16_4_int8  | Test Macro F1 | 0.689024     | 0.689601      | In-Memory Eval | Disk FlatBuffer | 0.000577      | 0.08%        | Mode 3: Quantization arithmetic rounding     |

calibration. Corrected by evaluating the finalized on-disk FlatBuffer directly ( $\Delta\text{Acc} \leq 3.26\%$ ).

- Defect Mode 2 (In-Memory vs. True Integer Arithmetic, 2 items): In-memory evaluation simulated quantization via fake-quantization operators rather than executing true integer arithmetic, producing a 7.82% Macro F1 discrepancy in mlp\_14f\_int8.
- Defect Mode 3 (Discretization Rounding, 8 items): Minor parameter rounding during FlatBuffer conversion ( $\Delta \leq 0.19\%$ ).

#### B. RQ2: Quantization Verification Results

Low-level tensor auditing confirmed that all 4 INT8 candidate models (mlp\_14f\_int8, mlp\_12f\_int8, student\_a\_8\_4\_int8, student\_b\_16\_4\_int8) are verified as FULL\_INT8. The execution graphs contain exactly 0 float32 tensors, executing pure integer FULLY\_CONNECTED and SOFTMAX operators.

#### C. RQ3: Sparsity vs. Storage Decoupling Findings

Visual inspection of pruned models revealed that while 75% magnitude pruning achieved 73.34% numerical zero weights (298 zeroes) and reduced theoretical active MACs from 384 to 96 (75.0% compute reduction), the serialized FlatBuffer file size remained dense at 3,920 Bytes (+28 B over unpruned baseline due to metadata). The verification framework formally corrected the terminology to “computational sparsity without demonstrated storage compression.”

#### D. RQ4: Threshold Optimization Leakage Bias

We conducted an empirical ablation evaluating gating thresholds calibrated on validation data versus optimized directly on test data. Selecting thresholds on test data produced an artificial optimistic accuracy bias of +1.80%, demonstrating that split-isolated calibration is mandatory to prevent inflated performance claims.

#### E. Physical On-Device Deployment Validation Case Study

To demonstrate the artifact verification protocol across the final deployment transformation boundary (FlatBuffer  $\rightarrow$  C-Header  $\rightarrow$  Bare-Metal MCU Binary), the four verified FULL\_INT8 candidate models were deployed to an ESP32-D0WD-V3 microcontroller (Xtensa LX6 dual-core @ 240 MHz, 4 MB Flash, 320 KB SRAM). Isolated single-sample execution latency was measured via esp\_timer\_get\_time() across 24,000 iterations (3 rounds  $\times$  2,000 iterations per model):

- student\_a\_8\_4\_int8 (176 params): ESP32 Mean = 64.55  $\mu\text{s}$ , P95 = 69.00  $\mu\text{s}$  (Host x86: 1.02  $\mu\text{s}$ , Slowdown: 63.3 $\times$ ).
- student\_b\_16\_4\_int8 (328 params): ESP32 Mean = 72.96  $\mu\text{s}$ , P95 = 83.00  $\mu\text{s}$  (Host x86: 0.98  $\mu\text{s}$ , Slowdown: 74.5 $\times$ ).
- mlp\_12f\_int8 (380 params): ESP32 Mean = 76.77  $\mu\text{s}$ , P95 = 83.00  $\mu\text{s}$  (Host x86: 1.00  $\mu\text{s}$ , Slowdown: 76.8 $\times$ ).
- mlp\_14f\_int8 (412 params): ESP32 Mean = 89.90  $\mu\text{s}$ , P95 = 95.00  $\mu\text{s}$  (Host x86: 1.43  $\mu\text{s}$ , Slowdown: 62.9 $\times$ ).

TABLE III  
The 5-Tier Evidence Classification Framework

| Evidence Tier | Epistemological Level         | Representative Project Metric                                       |
|---------------|-------------------------------|---------------------------------------------------------------------|
| Tier 1        | Direct Physical Measurement   | ESP32 latency (64.55 $\mu$ s), Disk size (3,892 B), Arena (916 B)   |
| Tier 2        | Host Empirical Measurement    | Host x86_64 single-sample latency (0.98 $\mu$ s)                    |
| Tier 3        | Reproducible Simulation       | 80-config QoS trace simulation (100% compliance)                    |
| Tier 4        | Derived Operational Estimate  | Expected MACs under 90% nominal prior (39.1)                        |
| Tier 5        | Literature-Supported / Future | Collinear feature reduction ( $r = 0.997$ ), on-vehicle CAN latency |

This physical deployment case study highlights two crucial methodological findings: (1) physical ESP32 execution provides an additional empirical deployment tier in the case study, establishing an observed host-to-ESP32 latency slowdown ratio between  $62.9\times$  and  $76.8\times$  without conflating host simulation timing with embedded execution; and (2) low-level memory inspection verified that TensorFlow Lite Micro MicroAllocator committed 916 Bytes of static working tensor memory with zero runtime heap allocations.

## VI. Evidence Tier Classification Framework

To guard against unsubstantiated claims in the edge AI literature, Table III establishes a 5-tier evidence hierarchy. Each metric reported in this work is strictly associated with its authoritative epistemological tier.

## VII. Scope and Generalization

### A. General Principles vs. Case Implementation

We explicitly distinguish general verification principles from our specific case implementation:

- General Principle: Deployment metrics should be evaluated directly on compiled disk binaries. Case Implementation: Evaluated on TensorFlow Lite FlatBuffers via Python C-API bindings.
- General Principle: Execution dtypes should be audited for zero floating-point fallback. Case Implementation: Parsed TFLite FlatBuffer tensor details.
- General Principle: Sparsity should be distinguished from storage compression. Case Implementation: Evaluated dense FlatBuffer representation of pruned MLPs.

## VIII. Threats to Validity

### A. Internal Validity

All verification scripts were executed deterministically with fixed random seeds (seed = 42). The audit pipeline parses exact binary headers using official TensorFlow Lite APIs.

### B. External Validity

Our case study evaluated a suite of multi-layer perceptron models for multi-sensor diagnostic classification. While convolutional and recurrent topologies exhibit different operator graphs, the principles of binary inspection, data isolation, and timing protocols generalize across all TinyML domains.

## IX. Limitations

- 1) Single Case Study Scope: Evaluated on one end-to-end multi-sensor diagnostic pipeline; multi-application cross-domain auditing remains future work.
- 2) Model Topology Focus: Evaluated on feedforward MLPs; CNNs and transformers require future parser extensions.
- 3) Runtime Specificity: Implemented for TensorFlow Lite FlatBuffers; alternative runtimes (microTVM, ONNX Runtime Mobile) are unverified.
- 4) Empirical Latency vs. Formal WCET: While physical single-sample latency was measured on 240 MHz ESP32 silicon (64.55–89.90  $\mu$ s), these reflect empirical on-device distributions and do not establish formal static Worst-Case Execution Time (WCET) bounds.
- 5) Energy Profiling Boundary: Physical power dissipation on silicon was not measured with hardware power analyzers.
- 6) Single Engine Domain: Data reflects physical dynamometer recordings from one engine platform.
- 7) Empirical Bias Demonstration: The +1.80% threshold leakage bias is specific to the evaluated multi-fidelity pipeline.

## X. Reproducibility and Artifact Availability

All verification scripts (scripts/phase4\_5\_verification.py), audited model FlatBuffers, physical ESP32 PlatformIO deployment firmware (phase5/firmware/), and discrepancy resolution logs are publicly archived in the repository.

## XI. Conclusion

This paper presented an artifact-driven empirical verification protocol and defect taxonomy for compiled TinyML deployment artifacts. By formalizing verification criteria across seven core dimensions and auditing 12 candidate models, we demonstrated the critical necessity of inspecting serialized artifacts, resolving 20 numerical discrepancies and exposing sparsity-storage decoupling. Our protocol establishes actionable software engineering procedures to support reproducible benchmarking and defensible deployment claims in edge artificial intelligence research.

## Acknowledgment

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